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RECEIVED 10 January 2026

REVISED 11 March 2026

ACCEPTED 20 March 2026

PUBLISHED 08 April 2026

CITATION

Anggraini CN, Rachmadania N,
Nugroho C, Wulandari A and
Rina N (2026) How students make sense
of ChatGPT in digital literacy education:
reflections on ethics, dependency,
emotion, and evaluation.
Front. Polit. Sci. 8:1784691.
doi: 10.3389/fpos.2026.1784691

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How students make sense of ChatGPT in digital literacy education: reflections on ethics, dependency, emotion, and evaluation

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Generative AI tools such as ChatGPT are rapidly entering higher education and reshaping how students access, justify, and evaluate knowledge, issues relevant to academic integrity and information governance. However, most studies focus on adoption, attitudes, or performance outcomes, and fewer examine how students reflectively make sense of ChatGPT after structured digital literacy instruction, including how ethics, dependence, emotion, and credibility evaluation intersect. This study analyzed 133 undergraduate self-reflection texts produced after a Digital Literacy course at an Indonesian university. We conducted a qualitative content analysis using NVivo, combining prompt-guided coding with iterative thematic abstraction to identify higher-order patterns across reflections. Three higher-order themes emerged, negotiated dependency captured situational reliance on ChatGPT alongside efforts to retain authorship and learning autonomy. Ethical ambivalence described boundary-setting around fairness, originality, and disclosure. Conditional trust reflected trust calibration through verification practices (e.g., cross-checking, selective rewriting) and recognition that fluent outputs may be unreliable. Emotional responses, including reassurance, anxiety, and guilt, often shaped the intensity of reliance and the likelihood of verification. Students did not uniformly treat ChatGPT as an epistemic authority; instead, they calibrated their responsibility and trust in the face of uncertainty. The study's novelty lies in conceptualizing AI literacy as "layered positioning" across dependency, ethics, affect, and evaluative judgment, showing how these dimensions co-occur in students' narratives rather than operate separately. The findings support policy and curriculum design that make disclosure expectations explicit, teach verification routines, and use reflective assessment to strengthen student agency in AI-mediated knowledge environments.

KEYWORDS

AI ethics, ChatGPT, digital literacy, emotional response, information evaluation, self-reflection, students' dependency

1 Introduction

Generative artificial intelligence (GenAI), including large language model (LLM)-based chatbots such as ChatGPT, has become increasingly visible in higher education as students use it to generate ideas, draft text, summarize materials, and seek on-demand explanations (Holmes et al., 2024; Ravšelj et al., 2025; Baig et al., 2024; Gonsalves, 2025). Recent reviews suggest that generative AI may offer pedagogical benefits in some contexts, including efficiency gains, support for idea generation, and expanded access to on-demand explanations, particularly for novice learners (Jensen et al., 2025; Kasneci et al., 2023; Abdallah et al., 2025; Klimova et al., 2025). At the same time, scholars caution that when AI outputs are adopted without scrutiny, students may become vulnerable to automation bias, over-trusting fluent system responses and under-detecting errors, especially when ChatGPT functions as an epistemic shortcut rather than a provisional draft requiring evaluation (Parasuraman and Manzey, 2010; Zhai et al., 2024). Accordingly, concerns about critical skills are best framed as conditional rather than deterministic: the risks are most pronounced when reliance is unregulated, while structured guidance and verification routines can support a more reflective, critical integration (Zhai et al., 2024; Gerlich, 2025; Tian and Zhang, 2025; Wang and Fan, 2025; Wang, 2024).

Prior work in the Digital Literacy course context at Telkom University found that some students reported difficulty disengaging from ChatGPT despite awareness of potential downsides for learning autonomy (Astuty and Anggraini, 2024). To situate this pattern conceptually, research increasingly frames intensive AI use as a form of cognitive offloading, delegating parts of interpretation and reasoning to an external system, which can be productive for scaffolding but may also weaken sustained engagement if not accompanied by monitoring and verification (Zhai et al., 2024; Tian and Zhang, 2025). Building on this perspective, the present study shifts from measuring use or attitudes to examining students' sense-making through reflective accounts, focusing on how reliance is negotiated alongside ethical reasoning, emotional experience, expressed concerns, and information-evaluation practices.

Educational responses to GenAI emphasize that outcomes depend not only on access to tools but also on pedagogical design, assessment expectations, and explicit guidance about disclosure and verification (Chiu et al., 2024; Fitrah, 2024; Sudirman et al., 2022; Christodoulou, 2025). Evidence from AI-in-education research indicates that teacher support and instructional framing can shape students' motivation and capacity to learn with AI-based chatbots (Chiu et al., 2024). In this context, Telkom University introduced a mandatory Digital Literacy course in the Communication Science program (2023/2024 academic year) designed to strengthen critical digital competence, including ethical awareness and credibility evaluation in AI-mediated information environments (Astuty and Anggraini, 2024). The course, therefore, provides an appropriate setting to examine how students interpret and regulate their use of ChatGPT after explicit instruction on responsible digital practice.

As part of the course assessment, students produced written self-reflections guided by prompts covering five areas: ethical considerations, dependency, emotional responses, concerns, and information evaluation. In the original manuscript, these areas could be read as descriptive categories. In this revision, we explicitly conceptualize them as operational dimensions of a theoretically grounded AI literacy framework. Specifically, (1) ethical considerations reflect the normative dimension emphasized in AI ethics and academic integrity

scholarship (Floridi et al., 2018; Eaton, 2023); (2) dependency reflects an epistemic dimension connected to automation bias and emerging accounts of epistemic dependency on AI systems (Parasuraman and Manzey, 2010; Ferrario et al., 2024); (3) emotional responses reflect an affective dimension of learning, consistent with control-value theory of achievement emotions, where affect shapes engagement and self-regulation (Pekrun, 2006; Park et al., 2023); (4) concerns capture a reflective-critical dimension linked to risk perception and trust calibration in human-AI interaction (Madsen and Gregor, 2024); and (5) information evaluation reflects a metacognitive-evaluative dimension aligned with media literacy and evaluative judgment frameworks that emphasize credibility assessment and cross-verification (Potter, 2019; Bearman et al., 2024; Carless and Boud, 2018). This framework provides an explicit rationale for examining these five areas together: they represent interrelated capacities required for responsible engagement with AI-generated information in educational settings.

Self-reflection is an appropriate method for examining these dimensions because it captures how learners interpret experiences, justify decisions, and articulate moral and epistemic reasoning within their own accounts (Brockbank and McGill, 2007; Potter, 2019). It is also particularly relevant in AI-mediated learning, where decisions about reliance, disclosure, and verification can be situational and shaped by perceived norms and uncertainty. Although large-scale indicators suggest that ChatGPT is widely adopted internationally and in Indonesia, the primary purpose of this study is not to restate adoption trends; rather, it is to understand how students describe the meaning of ChatGPT use in their academic workflow and how they navigate uncertainty about credibility and responsibility (Park et al., 2023; Jensen et al., 2025).

Existing studies have discussed ethical and integrity risks (e.g., originality and disclosure), the possibility of reduced critical engagement under unregulated reliance, and mixed emotional responses during chatbot interaction (Cotton et al., 2024; Eaton, 2023; Park et al., 2023; Kofinas et al., 2025; Sufendi and Kamal, 2023). However, fewer studies examine students' reflective narratives after structured digital literacy instruction that explicitly addresses ethical boundaries and verification norms. Reflective texts, therefore, provide a distinctive empirical window into how learners negotiate accountability, calibrate trust, and enact evaluation routines when generative AI becomes part of formal learning practice.

Accordingly, this study analyzes 133 undergraduate self-reflection texts to examine how students make sense of their use of ChatGPT across the five theoretically grounded dimensions described above. The study aims to (1) map how students position ChatGPT in relation to autonomy and authorship, (2) describe ethical boundary-making and disclosure reasoning, (3) identify affective experiences that shape reliance, (4) document concerns that inform trust calibration, and (5) examine reported practices of verification and evaluative judgment. By linking these dimensions within a coherent framework, the study contributes to AI literacy research by specifying how ethical, epistemic, affective, and evaluative orientations co-occur in students' accounts, and it offers implications for curriculum and policy design that strengthen responsible AI use in higher education.

2 Theoretical framework

The rapid integration of generative artificial intelligence (GenAI) into higher education requires a conceptual lens that moves beyond

descriptive accounts of student use toward a more structured understanding of AI literacy as a multidimensional competence. In this study, AI literacy is conceptualized as a reflective–epistemic competence that integrates normative, cognitive, affective, and evaluative dimensions. Rather than treating students’ reflections on ChatGPT as isolated themes, this research situates them within a theoretically informed framework that draws on AI ethics, epistemic dependency theory, affective learning theory, risk and trust in technology research, and media literacy scholarship.

First, the normative dimension of AI literacy concerns ethical awareness and academic integrity. Global AI ethics guidelines emphasize principles such as transparency, accountability, and responsibility in the development and use of AI systems (Floridi et al., 2018; Jobin et al., 2019; Kim, 2021; Saputra et al., 2023). Within higher education, these principles intersect with academic integrity frameworks that address disclosure, authorship, and responsible technological assistance (Eaton, 2023). Ethical awareness, therefore, is not limited to rule compliance but reflects students’ capacity to internalize normative standards governing AI use in academic contexts. In this study, students’ reflections on ethical boundaries, disclosure practices, and perceptions of acceptable assistance are interpreted as manifestations of this normative dimension of AI literacy.

Second, AI literacy includes an epistemic dimension related to dependency and cognitive positioning toward AI systems. Research on automation bias and cognitive offloading demonstrates that individuals may delegate reasoning processes to technological systems, sometimes reducing critical engagement (Parasuraman and Manzey, 2010; Zhai et al., 2024; Dzindolet et al., 2003). Ferrario et al. (2024) conceptualize this phenomenon as epistemic dependency, wherein users position AI as an authority in knowledge production. Within learning environments, such dependency can reshape students’ roles from active constructors of knowledge to consumers of algorithmically generated outputs. In this framework, students’ reported reliance on ChatGPT, particularly as a starting point for thinking or as a substitute for ideation, represents an epistemic positioning toward AI. Dependency is thus interpreted not merely as frequency of use but as a reflective indicator of how students negotiate authority, autonomy, and cognitive effort in AI-supported learning.

Third, AI literacy encompasses an affective dimension. According to the control-value theory of achievement emotions (Pekrun, 2006; Pekrun and Linnenbrink-Garcia, 2014), students’ emotional experiences influence motivation, engagement, and self-regulation. Emerging research on human–AI interaction further shows that AI systems can evoke both emotional comfort and ambivalence (Heung and Chiu, 2025; Madsen and Gregor, 2024). In educational settings, AI chatbots may function as emotional scaffolds that reduce anxiety while simultaneously generating tension related to trust and academic responsibility. Students’ reflections describing relief, confidence, anxiety, guilt, or frustration are therefore interpreted as indicators of the affective dimension of AI literacy (Pekrun, 2006; Park et al., 2023; D’Mello and Graesser, 2012; Gross, 2015). These emotional responses shape patterns of use and mediate the relationship between technological affordance and learning behavior.

Fourth, this framework incorporates a reflective-critical dimension captured through students’ expressed concerns. Trust in technology research highlights that user trust is dynamic and fragile, especially when systems demonstrate inconsistency or opacity (Sullivan and Kim, 2018; Volpato et al., 2025; Alzamil, 2025). In the context of generative AI, concerns about accuracy, hallucination, data privacy, and academic

consequences reflect users’ risk perception (Jensen et al., 2025). Rather than treating concern as a negative attitude, media literacy scholarship suggests that skepticism can function as a catalyst for critical evaluation (Potter, 2019). Students’ articulated worries about misinformation, plagiarism, and long-term cognitive impact are thus understood as reflective-critical awareness, signaling emerging capacities to interrogate the reliability and implications of AI systems.

Finally, AI literacy requires a metacognitive evaluative dimension. Media and information literacy frameworks emphasize the ability to assess credibility, validity, and source quality in digital environments (Potter, 2019; Jones-Jang et al., 2021; Metzger et al., 2010). In higher education, this evaluative capacity is further conceptualized as evaluative judgment, the ability to make reasoned judgments about the quality of work and information (Carless and Boud, 2018; Bearman et al., 2024). Within AI-mediated learning, evaluative judgment becomes crucial because generative systems can produce coherent yet inaccurate outputs. Students’ reported practices of cross-verification, source comparison, and selective adoption of AI-generated text are interpreted as manifestations of this metacognitive dimension. Conversely, uncritical acceptance of AI output reflects its relative absence.

Taken together, these five interrelated dimensions, normative (ethical awareness), epistemic (dependency), affective (emotional responses), reflective-critical (concerns), and metacognitive (information evaluation), constitute a reflective–epistemic AI literacy framework. This integrative perspective allows the study to analyze students’ self-reflections not as fragmented experiences but as interconnected expressions of how they negotiate responsibility, authority, emotion, risk, and judgment in AI-supported learning. The five analytic areas examined in this research are therefore theoretically grounded in established scholarship and represent operationalizations of broader AI literacy competencies within higher education contexts.

3 Methods

3.1 Research design and approach

This study employs a content analysis design to examine students’ self-reflection data. Content analysis refers to a systematic technique for identifying patterns, themes, and meanings within textual data through structured coding and categorization (Krippendorff, 2018; Creswell, 2014). This approach is particularly suitable for examining reflective narratives, as such narratives represent participants’ lived experiences, interpretations, and evaluative positioning regarding their use of ChatGPT. In addition to content analysis, the study incorporates thematic abstraction procedures to identify patterned meanings across student reflections. The analytic process moves from coding surface-level expressions to interpreting higher-order thematic patterns (Braun and Clarke, 2006).

3.2 Context and participants

This study was conducted within the Digital Literacy course in the Communication Science program at the Faculty of Communication and Social Sciences, Telkom University, Bandung, West Java, Indonesia during the 2023/2024 academic year. The course was designed to enhance students’ digital literacy competencies, including awareness of digital ethics, information evaluation skills, and the responsible use of

artificial intelligence technologies (Astuty and Anggraini, 2024). Participants were undergraduate first-semester students enrolled in the mandatory Digital Literacy course within the Communication Science program during the 2023/2024 academic year ($n = 133$). Students were selected as participants because they represent active users of AI technologies in academic contexts and demonstrate high levels of interaction with ChatGPT (Lukman and Aisy, 2023). The prior research within the same course identified patterns of optimization and emerging dependency in students' ChatGPT use (Astuty and Anggraini, 2024). The present study extends this empirical foundation by examining how such patterns are articulated reflectively across multiple dimensions.

3.3 Data collection methods

The research data were collected through written self-reflection assignments administered to students after completing the Digital Literacy course, particularly in the modules addressing the use of AI and ChatGPT. Data were collected exclusively through written self-reflection assignments. No interviews or focus groups were conducted for this study. Self-reflection was selected as the data collection method because it captures students' experiential accounts, emotional articulation, and evaluative reasoning in a personal and contextualized manner (Brockbank and McGill, 2007; Potter, 2019; Mezirow, 2000). The self-reflection instructions were designed to guide students in reflecting on their experiences with ChatGPT across five key dimensions: (1) ethical awareness of ChatGPT use, (2) levels of reliance and dependency, (3) emotional responses, (4) concerns regarding ChatGPT use, and (5) information evaluation skills related to ChatGPT-generated outputs. The reflective prompts were open-ended to allow students to freely express their perspectives and experiences without the constraints of predefined response.

3.4 Data analysis procedures

Data analysis was conducted using NVivo software to facilitate both the coding process and the management of qualitative data. The analytical procedure followed the stages of thematic analysis, including initial coding (open coding), grouping codes into thematic categories, and developing overarching themes that reflect patterns in students' reflections (Braun and Clarke, 2006). Data analysis followed an iterative qualitative content analysis, complemented by reflexive thematic analysis, to develop higher-order themes. The dataset comprised 133 written self-reflection texts produced after completion of a Digital Literacy course module on generative AI. Texts were de-identified before analysis and imported into NVivo for data management. The unit of analysis was meaning (a sentence or short segment expressing a single idea about ChatGPT use).

First, the researchers conducted corpus familiarization through full readings and analytic memoing to identify recurrent patterns and to document early interpretive questions (Braun and Clarke, 2006). Second, performed first-cycle open coding using an inductive approach, assigning descriptive codes to meaning units that captured students' stated practices, justifications, emotions, and evaluative actions. Third, the initial code set was consolidated by merging synonyms, refining definitions, and clarifying code boundaries to improve consistency. All data were handled responsibly and exclusively for scholarly purposes and improve consistency.

Fourth, codes were grouped into analytic categories using pattern/axial coding. While the self-reflection prompts (ethics, dependency,

emotion, concerns, and information evaluation) served as sensitizing dimensions for organizing early coding, they did not determine the final themes. Fifth, conducted cross-category comparison to identify relational patterns that cut across the prompt dimensions (e.g., how ethical concerns shaped verification, or how emotional reassurance intensified reliance). Sixth, through iterative abstraction, developed and refined candidate themes by checking internal coherence (whether excerpts within a theme fit together) and external distinction (whether themes were meaningfully different from each other). Finally, conducted a deliberate search for divergent and minority cases to avoid overgeneralization and to ensure that themes captured both dominant and contrasting positions. The analytic pathway from raw texts to codes, categories, and higher-order themes is summarized in Figure 1.

3.5 Trustworthiness and credibility

To ensure data trustworthiness, this study applied principles of credibility and transparency throughout the analytical process. Coding consistency was ensured through iterative reviews of the developed codes and categories. In addition, researcher reflexivity was addressed by acknowledging the researcher's positionality within the research context and potential sources of bias (Lincoln and Guba, 1985). Analytical decisions were continuously compared against raw data excerpts to ensure coherence between interpretation and evidence. The use of direct quotations from students' self-reflections in the Results and Discussion sections aims to strengthen the empirical grounding of the study's findings.

3.6 Ethical considerations

This study adhered to the principles of social research ethics appropriate to the context of higher education research. All student self-reflection data were collected as part of course-related learning activities and were used for research purposes only after the instructional process had concluded. Participants' identities were anonymized to safeguard confidentiality and privacy, and no personal information was included that could enable individual identification. Students were informed that their self-reflections might be used for academic research purposes without any impact on their academic evaluation. All data were handled responsibly and exclusively for scholarly purposes, in accordance with qualitative research ethics principles that emphasize participant protection, data confidentiality, and scientific integrity (Lincoln and Guba, 1985).

This study received ethical approval from the Telkom University Ethics Committee (Approval No. 357/LIT06/PPM-LIT/2024, dated 20 September 2024). Student reflections were produced as part of the course assessment. After the instructional process concluded and final grades were submitted, students were informed that their anonymized reflections could be used for research. Participation was voluntary, refusal carried no academic consequences, and all data were anonymized before analysis. Informed consent was obtained verbally for the research use of the anonymized reflection texts. Verbal consent was used to minimize the collection of personally identifying information.

4 Results and discussion

The analysis of 133 student self-reflection texts was conducted in two stages: a prompt-guided coding across five analytic dimensions, followed by cross-category abstraction. While the initial coding

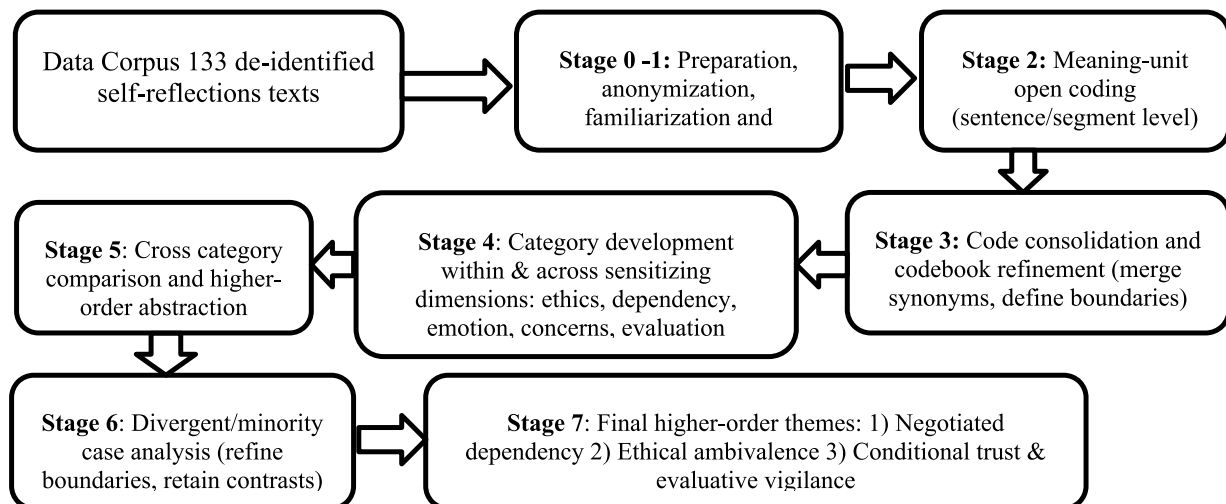


FIGURE 1
Theme development process (analytic pathway). Source: Authors' analysis.

organized responses around ethical awareness, dependency, emotional responses, concerns, and information evaluation, further iterative comparison revealed patterned relationships that extended beyond these discrete categories. Three higher-order themes emerged from the abstraction process: (1) Negotiated dependency; (2) Ethical ambivalence; and (3) Conditional trust and evaluative vigilance. These themes represent analytically derived configurations of meaning that capture how students position themselves in relation to ChatGPT within academic contexts. The findings indicate that students' reflections reflect interconnected orientations toward AI use. Many students articulated pragmatic reliance while simultaneously expressing evaluative awareness and emotional tension. At the same time, minority responses revealed more cautious or resistant stances, indicating variability rather than uniform adoption.

The following sections present each higher-order theme with representative excerpts and interpretive analysis. Results and discussion are integrated to ensure that interpretive claims remain grounded in students' narratives while being situated within relevant theoretical perspectives. To improve transparency, we report approximate frequency indicators for prominent patterns (based on coded segments across the corpus). These indicators are descriptive and intended to show how widely a pattern appears in the reflections, not to support statistical inference. Each theme section includes multiple excerpts (R01–R133) selected to represent dominant, variant, and divergent positions. Table 1 reports the approximate distribution of subthemes and provides an excerpt index, which improves traceability between claims and data.

4.1 Negotiated dependency

Across the corpus, dependency-related content appears in 78 of 133 reflections (items coded = 78), with 142 coding references in NVivo. Within this theme, students' positioning was not uniform: the most common pattern was managed reliance in which ChatGPT becomes part of the routine academic workflow, while students still articulate goals of control, autonomy, and eventual self-regulation (Faisal, 2024; Baker et al., 2024). The strongest indicator of deep reliance is the "belief" sub-dimension, 67 of 133 students expressed a conviction that they cannot stop using ChatGPT (belief items

coded = 67), followed by behavioral repetition (40 students) and emotional gratification (34 students).

Several students positioned ChatGPT as an initial support mechanism. One student wrote, "ChatGPT helps me start writing when I feel stuck. It gives structure and ideas, but I still revise it." Another explained, "It makes assignments faster and clearer, especially when I don't understand the topic." These accounts show that AI was used to reduce cognitive load at early stages of task completion. Students frequently framed ChatGPT as an efficiency and "starting point" tool, especially under time pressure or when they feel stuck (Acosta-Enriquez, 2024; McNamara et al., 2024). Informant 1 described a shift from occasional help to routine dependence: "I feel faster and more efficient asking ChatGPT directly than searching for manuals on Google. From then on, almost every time I had trouble understanding lecture material or was confused about structuring an argument, I would immediately open the AI." This expansion of use is paired with perceived cognitive difficulty when AI is removed: "I was confused about where to start and felt that my ideas weren't developed enough. At times like that, I realized that I had become very accustomed to, and even dependent on, the help of AI." At the more "committed" end of dependency, students stated an inability to stop using the tool: "I won't stop using ChatGPT"; "Once I discovered ChatGPT, I couldn't stop"; "I cannot..."

At the same time, some students expressed concern about increasing reliance (Budhathoki et al., 2024; Maulana and Darmawan, 2023). One student stated, "Sometimes I feel like I depend too much on it. I'm worried my critical thinking will decrease." Another reflected, "If I use ChatGPT too often, I become lazy to think deeply." These statements reveal awareness of potential cognitive offloading. Emotional responses also appeared to reinforce AI use. Some students described feelings of reassurance and reduced anxiety. One student wrote, "I feel calmer when I use ChatGPT because it reassures me that my answer is correct."

Despite these patterns, most students did not report replacing their own reasoning entirely. Several described modifying AI-generated output before submission. As one student explained, "I never copy it directly. I use it as reference and rewrite it in my own words." This practice reflects selective integration rather than direct adoption. Overall, the findings indicate that dependency on

TABLE 1 Summary of prevalence and illustrative excerpts.

Higher-order theme	Core indicators (NVivo)	Prevalence (Items coded/N = 133)	Illustrative excerpts
Negotiated dependency	Dependence node; beliefs/behavior/emotional sub-dimensions	Dependence 78/133; Beliefs 67/133; Behavior 40/133; Emotional 34/133	Informant; statements of inability to stop; willingness to stop; Informant: discontinuation
Ethical ambivalence	Integrity-risk reasoning; fear linked to plagiarism/accountability; authorship boundary rules	Fear 8/133 (integrity-risk expression)	Plagiarism fear; accountability of information; Informant reprimand; Informant: non-copying rule
Conditional trust and evaluative vigilance	Evaluation node; correctness/accuracy/appropriateness/how-it-works; verification norms	Evaluation 128/133; Correctness 122/133; Accuracy 50/133; Appropriateness 43/133; How it works 26/133; minority: 11 weak evaluations.	Correctness skepticism; validity caution; not sole source; Informant: error + rule; minority weak evaluation

Source: NVivo analysis (authors' data).

ChatGPT was not absolute. Students described patterns of use that involved assistance, monitoring, revision, and at times restraint. AI was integrated into learning practices while students continued to assert responsibility for comprehension and authorship. Negotiated Dependency therefore reflects managed reliance shaped by cognitive judgment and situational demands. However, the decision to rely on AI also introduced questions about authorship, fairness, and responsibility, which became more explicit in students' ethical reflections.

4.2 Ethical ambivalence

Even though the theme appeared in a smaller subset of reflections, it reveals an important dimension of AI literacy related to moral reasoning and academic integrity. Qualitative thematic analysis does not rely solely on frequency but also considers conceptual relevance and explanatory value (Braun and Clarke, 2006). Ethical ambivalence in this dataset is expressed less as abstract moral theory and more as practical risk reasoning: fear of plagiarism, fear of inaccurate outputs being used in academic work, and authorship concerns ("copying" vs. "genuine understanding"). This ethical orientation is visible in the emotion node "fear" (8 students), where fear is explicitly linked to plagiarism and accountability of information. In other words, students' ethics talk is often triggered by perceived consequences, being accused of plagiarism, being evaluated by lecturers, or being held responsible for wrong information.

Students' reflections show sustained moral positioning in relation to ChatGPT use. Many accounts focused on questions of permissibility, authorship, and disclosure. One student wrote, "I'm afraid that using ChatGPT too much could be considered cheating." Another stated, "Sometimes I wonder whether it's still my work if I rely on AI." These statements indicate that students evaluated AI assistance against internalized standards of academic integrity. Concerns about originality and proper attribution were central to this evaluation. Other students justified their use of ChatGPT within what they perceived as acceptable limits. One reflection stated, "As long as I understand the material and don't copy directly, I think it's still ethical." Another wrote, "I use it for ideas, but the final answer is mine." These accounts reveal attempts to define boundaries between assistance and substitution. Ethical judgment was framed as context-dependent and related to degree of reliance. This boundary-setting process reflects the practical negotiation of integrity standards rather than simple rule compliance.

Several excerpts clearly show this moral calibration. Students described fear of crossing integrity boundaries: "The emotion I felt was fear, because I was afraid of committing plagiarism or giving the wrong answer." This anxiety also appears as accountability concern: "The emotion I felt was fear... whether the information provided by the AI could be trusted." Relatedly, another informant described an incident where AI-supported work produced negative academic consequences: "I tried using the references suggested by the AI, but it turned out to be wrong and I got scolded by the lecturer. That was embarrassing and a bit traumatic." This experience is not just about accuracy; it becomes an ethical lesson about responsibility: "I feel disappointed because I trusted the answers too much... it turns out that they still need to be sorted and reviewed."

Ethical concerns were not limited to cheating or authorship (Khairul et al., 2023; Masrichah, 2023). Several students questioned the neutrality of AI-generated content. One student noted, "ChatGPT might give biased answers without us realizing it." Another commented, "It sounds convincing, but we don't know whose perspective it represents." These reflections indicate awareness that AI systems embed values and assumptions derived from training data. Importantly, ambivalence in these reflections did not indicate confusion alone. Many students articulated simultaneous recognition of AI's usefulness and its ethical complexity. One student wrote, "It helps me a lot, but I still feel unsure if I should rely on it." Another explained, "I think it depends on how responsibly we use it." These responses demonstrate reflective judgment.

Ethical Ambivalence therefore reflects an ongoing process of moral calibration. This moral evaluation was closely connected to how students assessed the reliability of AI outputs and determined the level of trust they were willing to extend. Students did not treat AI use as inherently legitimate or illegitimate. Instead, they evaluated it through concerns about authorship, fairness, bias, and long-term intellectual development. Their reflections indicate engagement with integrity norms, recognition of AI's embedded values, and context-based ethical judgment.

4.3 Conditional trust and evaluative vigilance

This theme is strongly supported by NVivo evidence on evaluation practices. In the evaluation node, 128 of 133 students were coded as engaging some form of evaluative reasoning (items coded = 128), with

245 coding references. Within this, correctness judgments are dominant (122 students), followed by attention to accuracy (50 students), appropriateness/fit (43 students), and how ChatGPT works (26 students). This distribution supports the interpretation that trust is often conditional: students may use ChatGPT, but many simultaneously articulate the need to test truth value, contextual fit, and limitations.

Some students expressed confidence in the system's usefulness. One student wrote, *"I trust ChatGPT to give me a good starting point."* Another stated, *"Most of the time the answers are clear and helpful."* These responses indicate that AI was perceived as a reliable resource for initiating or structuring academic tasks. At the same time, many students emphasized the need for verification. One student wrote, *"I always check again because ChatGPT can be wrong."* Another explained, *"It sounds convincing, but I still compare it with other sources."* These statements demonstrate awareness of epistemic limits.

Verification practices described by students included cross-checking with textbooks, consulting journal articles, and comparing answers with other online sources. One student stated, *"After using ChatGPT, I search the topic again in academic journals."* Another wrote, *"I don't copy it directly. I read it, then rewrite it after checking other references."* In this study, students' evaluative practices indicate that AI outputs were treated as provisional rather than authoritative. Skepticism in these reflections did not lead to rejection of the technology. Instead, students described regulating their reliance. One student wrote, *"I use it carefully and only as support."* Another explained, *"It's helpful, but I decide what to keep."* These statements show active management of information processing.

Students repeatedly stressed that AI outputs cannot be accepted as final truth (Kumar et al., 2024; Miller and Bartlett, 2024). Examples of correctness skepticism include: *"ChatGPT does not always provide the correct answer"; "In my opinion, ChatGPT sometimes provides inaccurate answers."* Accuracy vigilance appears in statements such as: *"Answers from ChatGPT are not always valid and must always be examined more deeply with the help of other sources..."* and *"The accuracy of ChatGPT's answers needs to be taken with a grain of salt and should not be the sole source of information."* Appropriateness/fit concerns, central to conditional trust, are voiced as: *"Sometimes, the answers provided seem arbitrary or not quite relevant to the questions asked."*

Divergent/minority pattern: despite the dominance of evaluative vigilance, the dataset also records a small subset with weaker evaluation. The file notes that 11 students did not conduct "true/false" evaluation adequately, indicating that conditional trust is uneven across the cohort. This supports a key interpretive point: evaluation practices are widespread but not universal, and a minority remain more vulnerable to over-trust or uncritical adoption.

5 Discussion

5.1 AI literacy as layered positioning

The results demonstrate that dependency is widespread but not uniform. In NVivo outputs, dependency-related content appears in 78/133 reflections, with belief-level statements (difficulty stopping) present in 67/133, alongside behavioral repetition (40/133) and emotional gratification (34/133). This distribution suggests that "dependency" in GenAI learning is not reducible to frequency of use. Rather,

it reflects epistemic positioning, how learners locate AI within their cognitive workflow and how far they allow it to substitute for reasoning steps. This aligns with epistemic dependency theory, which frames dependence as a relationship of epistemic authority and reliance rather than mere tool adoption (Ferrario et al., 2024; Baker et al., 2024). It also resonates with automation bias research, which shows that fluent, seemingly competent systems can become default epistemic anchors when users are pressed for time or when tasks are ambiguous (Parasuraman and Manzey, 2010).

Across 133 self-reflection texts, students' accounts show that AI engagement is not best described as either "useful tool" or "cheating shortcut." Instead, it is a layered positioning in which learners negotiate: (i) epistemic delegation (when and how to rely), (ii) normative boundaries (authorship, fairness, disclosure), and (iii) metacognitive evaluation (verification, source comparison, and decision rules). This pattern supports scholarship that defines AI literacy as a multidimensional competence rather than technical proficiency alone (Long and Magerko, 2020; Holmes et al., 2024). It also extends early higher education syntheses that caution against over-general claims about GenAI's impact, emphasizing instead that consequences depend on how learners regulate use, how tasks are assessed, and whether verification norms are taught and practiced (Jensen et al., 2025; Kasneci et al., 2023).

The findings show that AI engagement is structured through cognitive negotiation, ethical evaluation, and metacognitive regulation (Schraw et al., 2023; Panadero et al., 2023). These dimensions align with emerging scholarship that frames AI literacy as a multidimensional competence rather than a technical skill (Long and Magerko, 2020; Holmes et al., 2024; Tsang, 2025). The findings extend prior research on student dependency on generative AI. While earlier studies document risks of over-reliance and cognitive offloading (Zhai et al., 2024; Gerlich, 2025; Salido et al., 2025), the present data indicate that reliance is negotiated. Students described using ChatGPT to initiate tasks while retaining responsibility for revision and comprehension. This pattern is consistent with epistemic dependency theory (Ferrario et al., 2024) and automation bias research (Parasuraman and Manzey, 2010), yet it also demonstrates ongoing monitoring rather than passive surrender.

Compared to Astuty and Anggraini (2024), which identified patterns of optimization and difficulty disengaging, this study shows that dependency coexists with reflexive hesitation and boundary-setting. Students' awareness of potential cognitive decline aligns with broader concerns in higher education about generative AI and critical thinking (Jensen et al., 2025; Lo, 2024; Gerlich, 2025; Wang and Fan, 2025). However, the presence of self-monitoring suggests that literacy development may mediate these risks.

5.2 Ethical ambivalence as reflexive moral positioning

The second theme shows that ethical reasoning primarily emerges as practical moral calibration rather than as abstract ethical theory. Students repeatedly evaluate whether AI assistance crosses into plagiarism, whether outputs can be "owned," and whether use should be disclosed. This maps directly onto the normative dimension of the reflective-epistemic framework and aligns with academic integrity scholarship arguing that GenAI shifts integrity from simple prohibition toward boundary-making about authorship, transparency, and acceptable assistance (Eaton, 2023; Cotton et al., 2024). Students'

uncertainty about “my work” versus “AI work” indicates that integrity norms are being renegotiated at the learner level rather than imposed institutionally.

Students’ uncertainty regarding authorship and disclosure reflects internalized integrity norms rather than ignorance of them. This aligns with governance-oriented AI ethics frameworks that emphasize accountability and responsible use (Floridi et al., 2018; Jobin et al., 2019). The data further indicate that students recognized AI systems as value-laden rather than neutral tools. Ethical concerns extended to bias, opacity, and representational imbalance. This resonates with discussions of embedded moral assumptions in AI systems (Kim, 2021). Within higher education, such recognition reflects early stages of ethical AI literacy (Gandasari et al., 2024). Educational theory positions reflective judgment as central to civic literacy (Zimmerman, 2002). In this study, ambivalence functioned as reflective engagement rather than indecision. Students’ moral evaluation demonstrates interpretive effort in response to shifting technological norms.

The dataset also shows that ethical ambivalence is closely linked to risk and accountability. Students’ fear is often less about AI itself and more about consequences (e.g., being sanctioned for plagiarism or being “responsible” for wrong information). This fits governance-oriented AI ethics frameworks that emphasize accountability and responsibility in human–AI systems (Floridi et al., 2018; Jobin et al., 2019). It also supports a political-science-relevant interpretation: students treat ChatGPT use as a problem of responsibility allocation in an AI-mediated knowledge environment, who is accountable for outputs, how authority is justified, and what counts as legitimate knowledge work.

Ethical ambivalence also includes recognition that AI outputs are not neutral. Students raise concerns about bias, opacity, and “whose perspective” is represented. This complements global analyses of AI ethics guidelines that identify bias, transparency, and accountability as central concerns across governance regimes (Jobin et al., 2019) and reinforces that ethical AI literacy includes the capacity to interrogate system-level value imprints, not only individual cheating behavior. In practical terms, the evidence suggests that ethics instruction should not be confined to “do not plagiarize.” Still, it should explicitly address responsible epistemic conduct: disclosure expectations, authorship boundaries, and accountability for verifying claims generated by AI.

5.3 Conditional trust and metacognitive regulation

The evaluation evidence strongly supports the third theme. NVivo outputs indicate evaluation practices in 128/133 reflections, with correctness judgments dominant (122/133), followed by attention to accuracy (50/133) and appropriateness/fit (43/133). This distribution shows that students do not uniformly treat ChatGPT as an unquestioned authority; rather, many engage in trust calibration, a hallmark of conditional trust. This aligns with technology trust research that defines trust as dynamic and performance-based rather than stable or unconditional (Lankton et al., 2015; Sullivan and Kim, 2018).

The finding aligns with technology trust models that describe trust as shaped by perceived competence and reliability (Sullivan and Kim, 2018; Lankton et al., 2015). Students’ repeated emphasis on checking and cross-verifying indicates awareness of generative AI’s epistemic limitations. Empirical studies have documented hallucination rates and fabricated references in large language models (Chelli

et al., 2024). The verification practices described by students correspond to evaluative judgment frameworks in higher education (Carless and Boud, 2018; Bearman et al., 2024). Media literacy research similarly emphasizes cross-checking and source comparison as safeguards against misinformation (Jones-Jang et al., 2021; Potter, 2019).

Students’ insistence on verification corresponds to the epistemic reality that LLMs can produce fluent but incorrect outputs and fabricated references (Chelli et al., 2024; Zizka, 2025). Empirical work documenting hallucination and reference accuracy problems underscores why verification is not optional in AI-mediated learning environments (Chelli et al., 2024). In our data, verification is expressed as cross-checking with textbooks, journals, and other sources, reflecting the metacognitive-evaluative dimension of the framework. This directly connects to evaluative judgment scholarship: students must judge the quality of information and decide what to accept, revise, or reject (Carless and Boud, 2018; Bearman et al., 2024). Media literacy research likewise frames cross-verification and source comparison as core to credibility evaluation in digital information environments (Jones-Jang et al., 2021; Potter, 2019).

Across themes, students rarely positioned ChatGPT as an unquestioned authority. Instead, they described a cycle of reliance, ethical boundary-setting, and trust calibration through evaluation. Negotiated dependency explains when students delegate cognitive work; ethical ambivalence explains how they judge the legitimacy of that delegation; and conditional trust explains how they manage epistemic risk through verification. This integrated pattern supports the interpretation of AI literacy as layered positioning across epistemic, normative, affective, and evaluative dimensions.

6 Conclusion

This study examined how students interpret and regulate their engagement with generative AI within a Digital Literacy course. The findings indicate that student interaction with ChatGPT is structured through negotiated cognitive reliance, ethical ambivalence, affective responses, and conditional trust. These dimensions do not function in isolation: decisions about delegation were shaped by concerns about authorship and responsibility; emotions such as reassurance or hesitation influenced patterns of reliance; and trust was maintained through verification and selective integration. Together, the results contribute to debates on whether generative AI weakens or supports higher-order thinking in higher education by showing that students often recognize the risks of cognitive offloading and attempt to regulate their engagement through literacy practices rather than treating AI as an unquestioned authority.

By situating student reflections within epistemic, normative, affective, and evaluative dimensions, this study advances a multidimensional account of AI literacy. AI literacy is not limited to technical proficiency or rule compliance; it involves judgment under epistemic uncertainty, negotiation of integrity norms, emotional calibration, and ongoing evaluation of system outputs. The findings suggest that instructional responses should move beyond binary approaches of prohibition or unrestricted acceptance. Structured integration, through explicit disclosure expectations, guided verification routines, and reflective assignments, may strengthen students’

capacity to use AI responsibly while sustaining intellectual agency (Weng, 2024; Kofinas et al., 2025; Akiba, 2025).

This study has several limitations that should be considered when interpreting the findings. First, the qualitative analysis was conducted primarily by a single researcher, which may introduce interpretive bias in coding decisions, category boundaries, and theme naming. Although we maintained an audit trail (analytic memos, iterative code refinement, and checks against raw excerpts), future studies should strengthen confirmability through intercoder comparison, peer debriefing, or an external audit of the codebook and theme map. Second, the dataset consists of course-based self-reflection texts. Because reflections were produced within an academic context, responses may be influenced by social desirability or perceived expectations about “responsible” AI use, even when participation was voluntary and anonymized. Third, the sample is drawn from a single cohort of first-semester Communication Science students at one institution, within a course that explicitly teaches ethics and verification norms. This context likely shaped students’ language and evaluative practices, limiting transferability to other disciplines, academic levels, or institutions with different policy environments and assessment cultures. Finally, the study relies on self-reported accounts rather than direct behavioral measures; reflective awareness does not necessarily translate into sustained practice.

Future research should therefore triangulate reflective narratives with complementary data sources, such as interviews, writing drafts, citation trails, prompt logs (where ethically permissible), and instructor assessment artifacts, to examine how regulation practices are enacted in real coursework. Multi-site and cross-disciplinary studies would clarify whether the observed patterns generalize beyond this course context, and longitudinal designs could assess whether negotiated dependency and verification practices stabilize, intensify, or change as generative AI becomes more deeply embedded in academic infrastructures.

Data availability statement

The original contributions presented in the study are included in the article/supplementary material, further inquiries can be directed to the corresponding author.

Ethics statement

The studies involving humans were approved by The Telkom University Bandung Research Ethics Committee. The studies were conducted in accordance with the local legislation and institutional requirements. Written informed consent for participation was not required from the participants or the participants’ legal guardians/next of kin in accordance with the national legislation and institutional requirements.

Author contributions

CA: Conceptualization, Data curation, Funding acquisition, Investigation, Resources, Writing – original draft, Writing – review &

editing. NRa: Data curation, Formal analysis, Software, Writing – original draft. CN: Methodology, Supervision, Validation, Writing – review & editing. AW: Investigation, Resources, Visualization, Writing – review & editing. NRi: Project administration, Software, Visualization, Writing – original draft.

Funding

The author(s) declared that financial support was received for this work and/or its publication. This research was funded by the Directorate of Research and Community Service, Telkom University.

Acknowledgments

The authors express sincere gratitude to the students of the Communication Studies Program at Telkom University who participated in this research, and to the teaching team of the Digital Literacy course for their support in implementing the learning activities and facilitating data collection. Appreciation is also extended to the Directorate of Research and Community Service at Telkom University for its funding support. In addition, the authors thank the Digital Culture and New Media Studies Laboratory for its assistance in disseminating this research. The authors further acknowledge the Center of Excellence for Technological Society, Research Institute for Intelligent Business and Sustainable Economy, Telkom University, for providing research facilities.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was used in the creation of this manuscript. NVivo was used as the thematic analysis tool for open coding, categorization, and theme development. SciSpace was used to assist the literature review process. ChatGPT was used solely to support language editing and improve clarity of expression, while DeepL was used for proofreading. The authors declare that generative AI tools were not used for data analysis, interpretation, or the generation of scientific content. All intellectual contributions, analyses, and conclusions are the sole responsibility of the authors.

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